In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [5]:
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder , StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score,confusion_matrix,classificatio
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
```

In [7]:
```
df = pd.read_csv("Mall_Customers.csv")
df.head()
```

Out[7]:

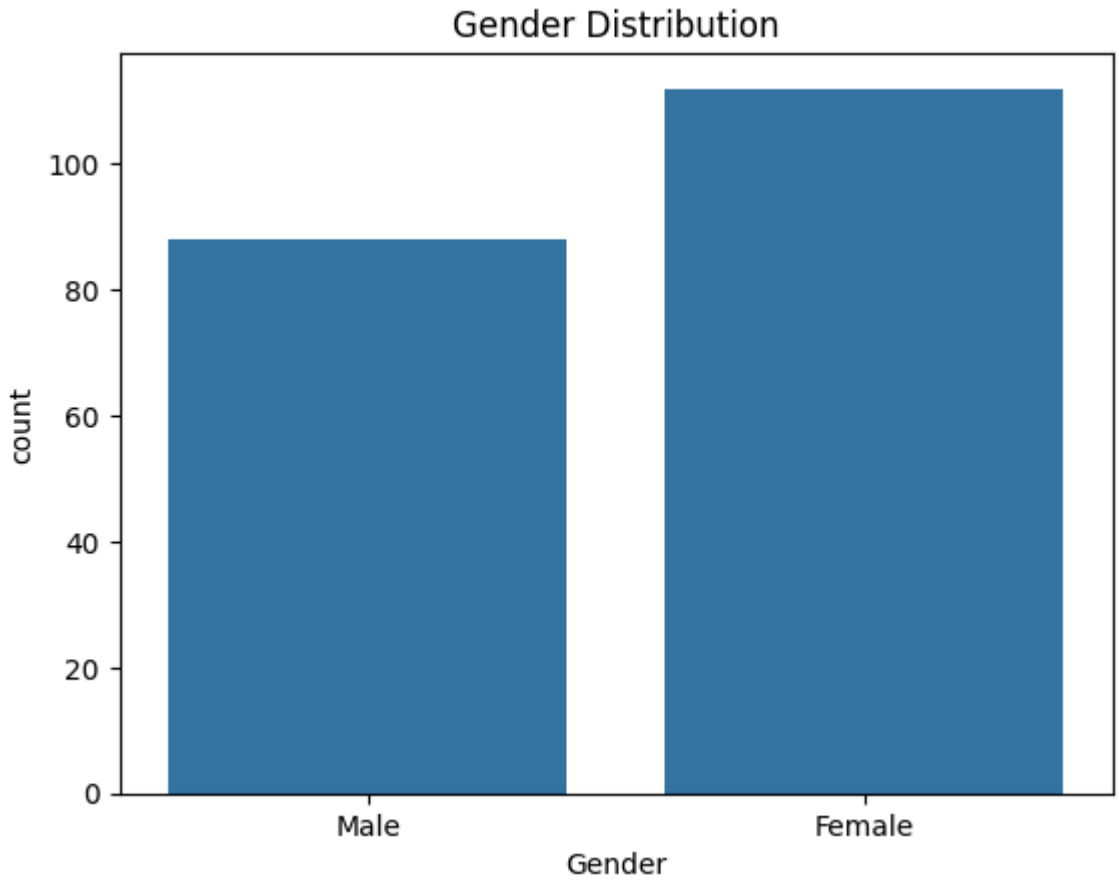
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [8]:
```
df.info()
df.describe()
df.isnull().sum()
```

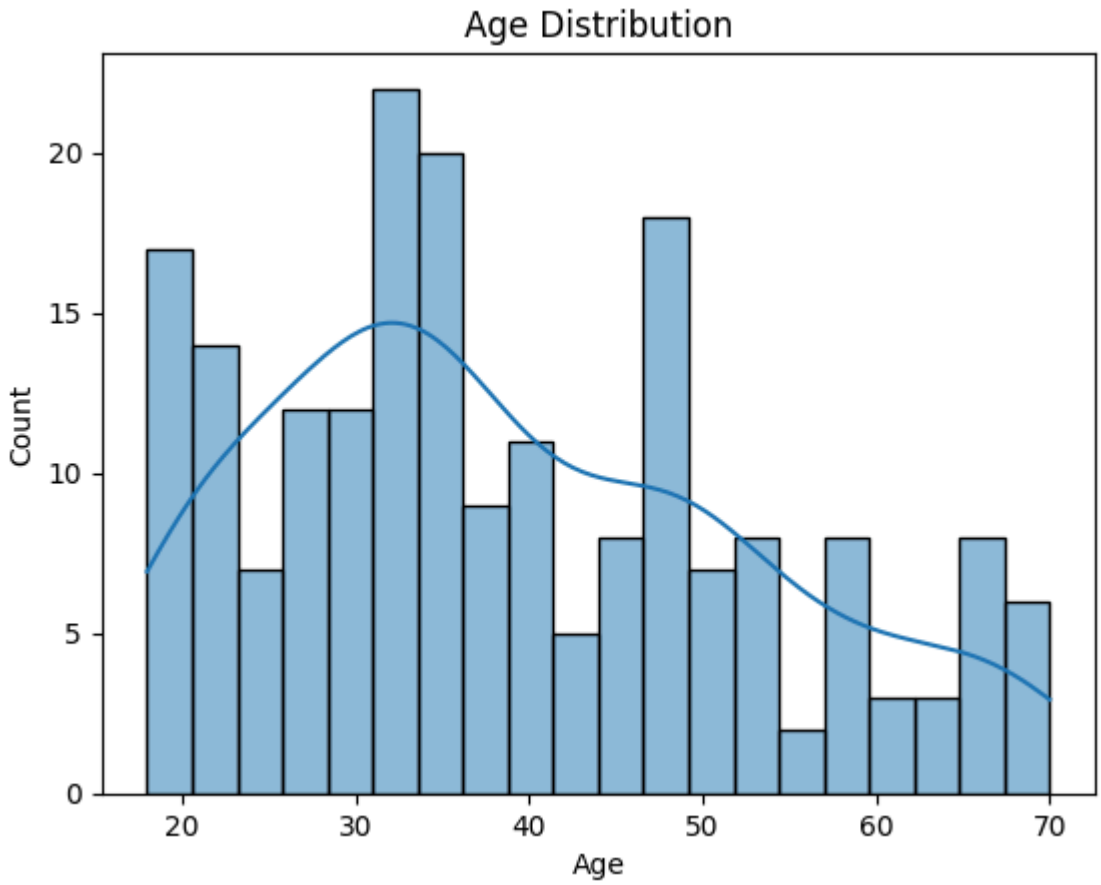
```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype 
---  ------                  --------------  ----- 
 0   CustomerID              200 non-null    int64 
 1   Gender                  200 non-null    str   
 2   Age                     200 non-null    int64 
 3   Annual Income (k$)      200 non-null    int64 
 4   Spending Score (1-100)  200 non-null    int64 
dtypes: int64(4), str(1)
memory usage: 7.9 KB
```

Out[8]:
```
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64
```

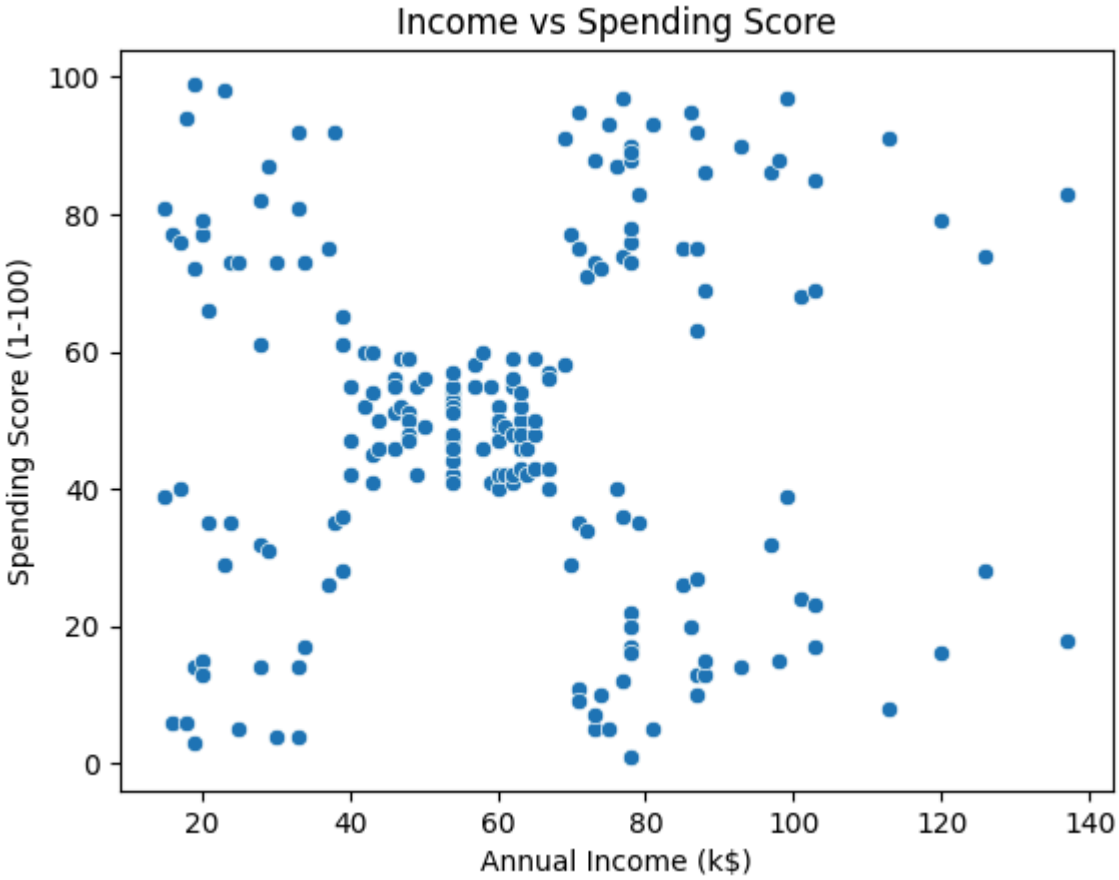
In [9]:
```
sns.countplot(x='Gender',data=df)
plt.title("Gender Distribution")
plt.show()
```



In [12]:
```
sns.histplot(df['Age'] , bins=20,kde=True)
plt.title("Age Distribution")
plt.show()
```



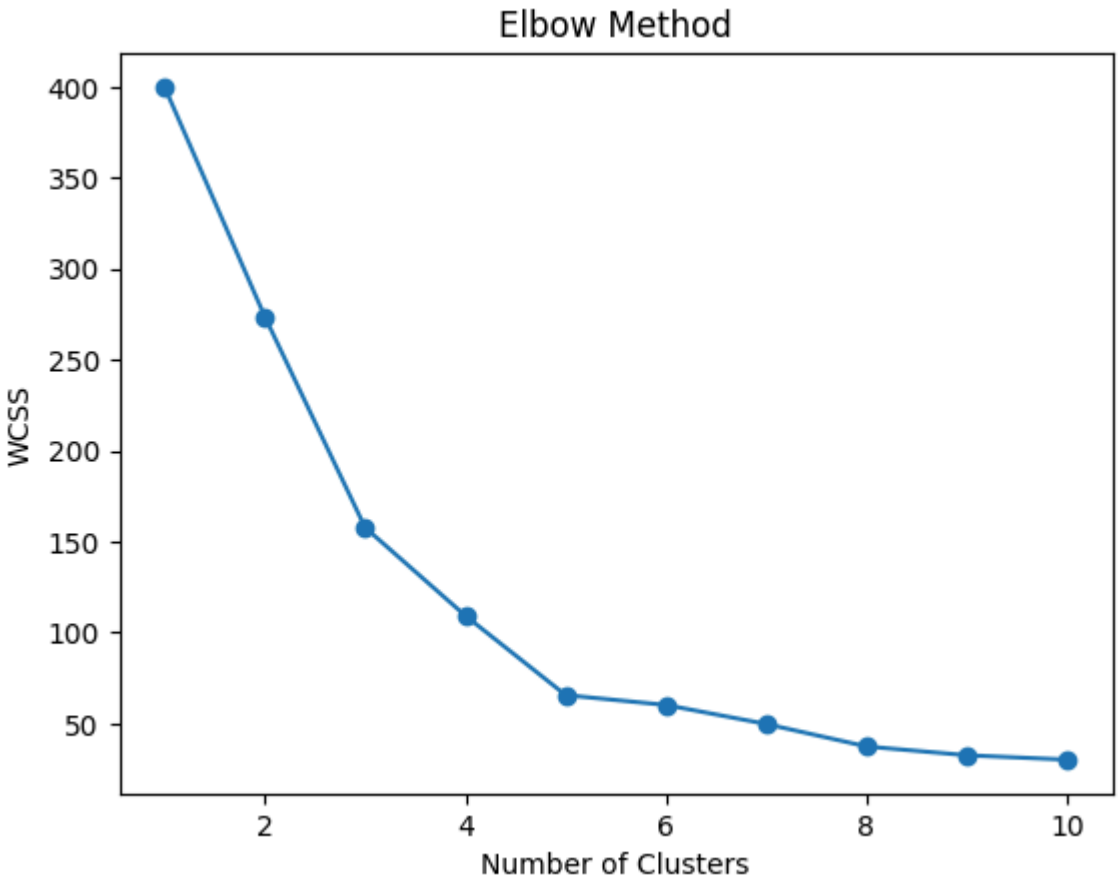
In [17]:
```
sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)', data=
plt.title("Income vs Spending Score")
plt.show()
```



In [18]:
```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

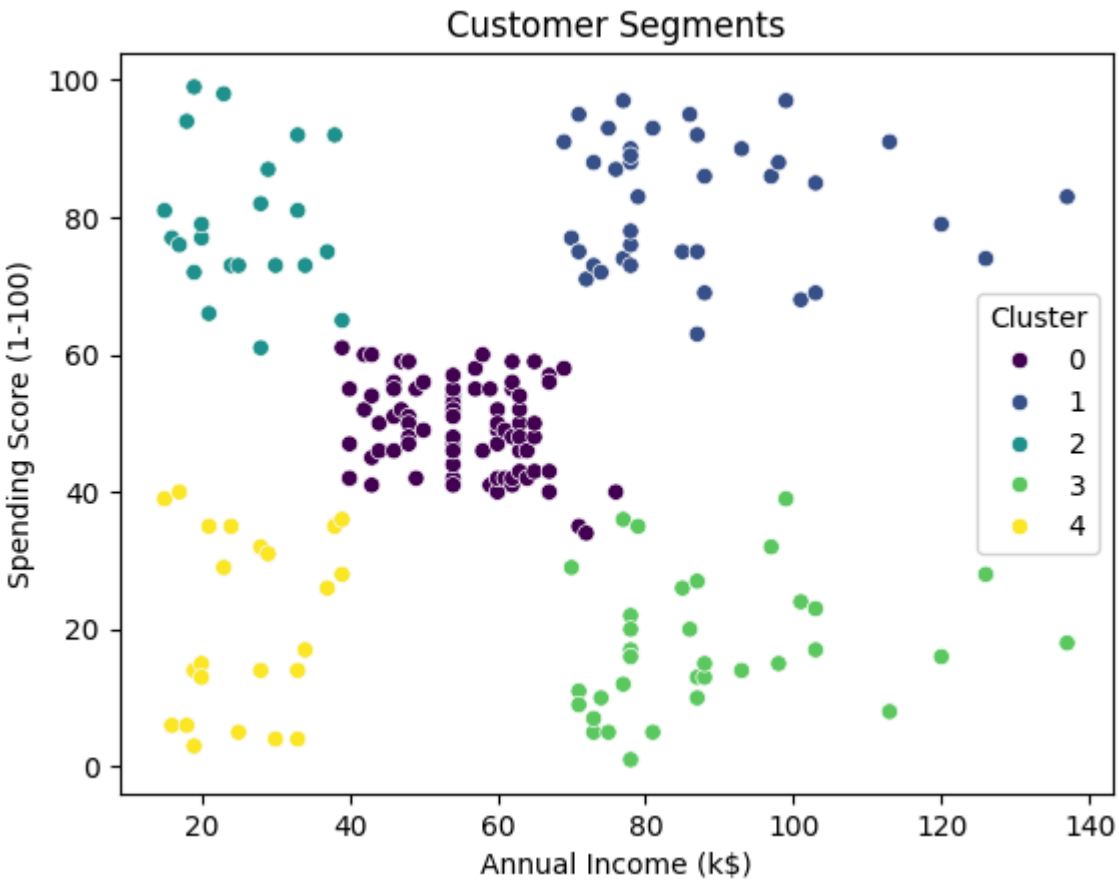
In [19]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [21]:
```
wcss = []

for i in range(1,11):
    kmean = KMeans(n_clusters=i,random_state=42)
    kmean.fit(X_scaled)
    wcss.append(kmean.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

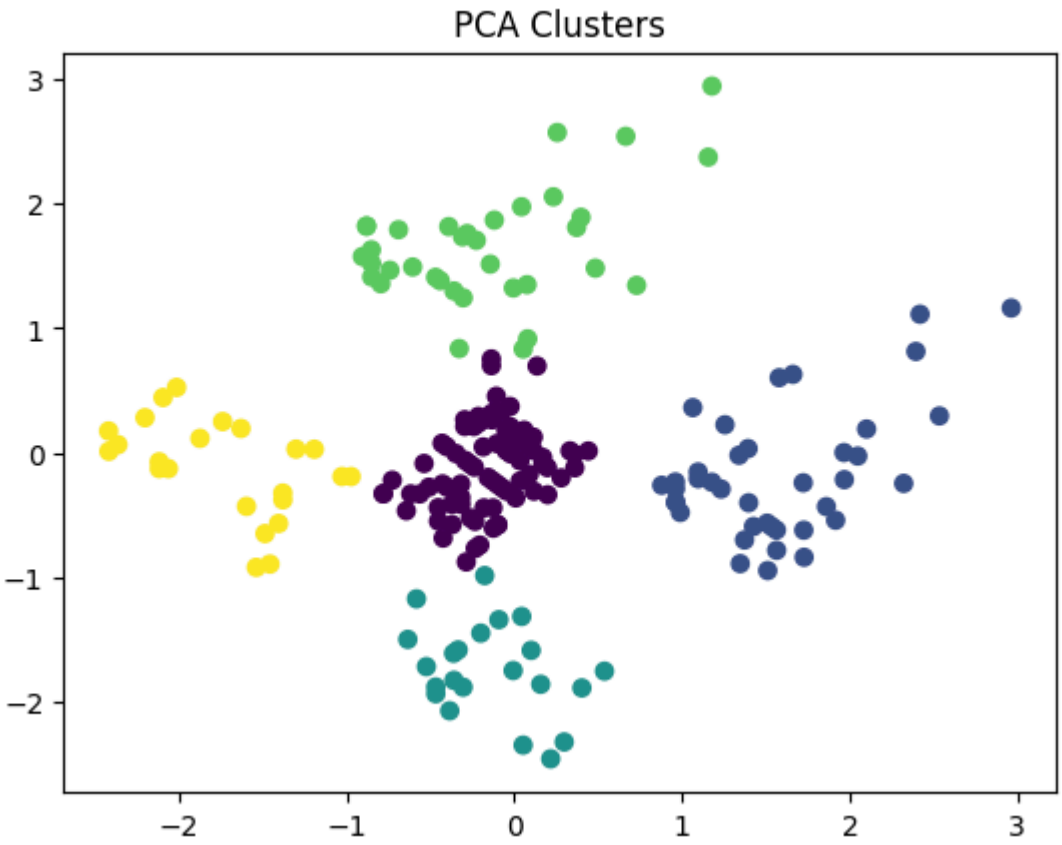


In [22]:
```
kmeans = KMeans(n_clusters=5,random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

In [23]:
```
sns.scatterplot(
    x='Annual Income (k$)',
    y='Spending Score (1-100)',
    hue='Cluster',
    palette='viridis',
    data=df
)

plt.title("Customer Segments")
plt.show()
```



In [24]:
```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

plt.scatter(X_pca[:, 0], X_pca[:, 1], c=df['Cluster'])
plt.title("PCA Clusters")
plt.show()
```



In [ ]: